Calcurion Physics Worksheets

A-Level Physics

Topic: Mechanics & Materials

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. State Newton's second law in terms of momentum. **[2]**

2. A force of 12 N acts for 0.5 s on a 3 kg mass initially at rest. Calculate the final velocity. **[3]**

3. Define the moment of a force. **[1]**

4. A 25 N force acts perpendicular to a 0.4 m lever arm. Calculate the moment. **[2]**

5. State the principle of moments. **[2]**

6. Define density and give its SI unit. **[2]**

7. An object of mass 0.8 kg has volume 2×10^{-4} m^3. Calculate its density. **[2]**

8. State Hooke's law. **[2]**

9. A spring of spring constant 150 N/m is extended by 40 mm. Calculate the force and the elastic energy stored. **[4]**

10. Define tensile stress and tensile strain. **[2]**

11. A wire of cross-sectional area 1.5×10^{-6} m^2 supports a load of 30 N. Calculate the stress. **[2]**

12. Define the Young modulus. **[1]**

13. A wire of original length 2.0 m extends by 1.5 mm under a stress of 1.2×10^8 Pa. Calculate the Young modulus if the strain is 7.5×10^{-4}. **[3]**

14. What is meant by the elastic limit? **[1]**

15. Sketch a stress–strain graph for a ductile material, labelling the elastic region and ultimate tensile strength. **[3]**

Section B – Hard (Q16–25)

16. A 2 kg mass moving at 4 m/s collides and sticks to a 3 kg mass at rest. Calculate the common velocity. **[3]**

17. A uniform beam of weight 120 N and length 2.0 m is pivoted at one end and held horizontal by a vertical cable at the other end. Calculate the tension in the cable. **[3]**

18. Derive $E = \frac{1}{2}F\Delta L$ for elastic energy from the force–extension graph. **[3]**

19. A steel wire of length 1.5 m and diameter 0.80 mm has Young modulus 2.0×10^{11} Pa. Calculate the extension under a 50 N load. **[5]**

20. Explain the difference between plastic and elastic deformation. **[3]**

21. A material has breaking stress 4.0×10^8 Pa. What maximum load can a wire of diameter 1.0 mm support? **[3]**

22. A car of mass 1200 kg decelerates from 25 m/s to rest in 4.0 s. Calculate the average braking force. **[3]**

23. State the conditions for equilibrium of a rigid body. **[2]**

24. A spring compresses by 5.0 cm when a 2.0 kg mass is placed on it. Calculate the spring constant and the energy stored. Take $g = 9.8$ N/kg. **[4]**

25. Describe an experiment to determine the Young modulus of a metal wire. **[5]**

End of Worksheet • *Show all working clearly.*